



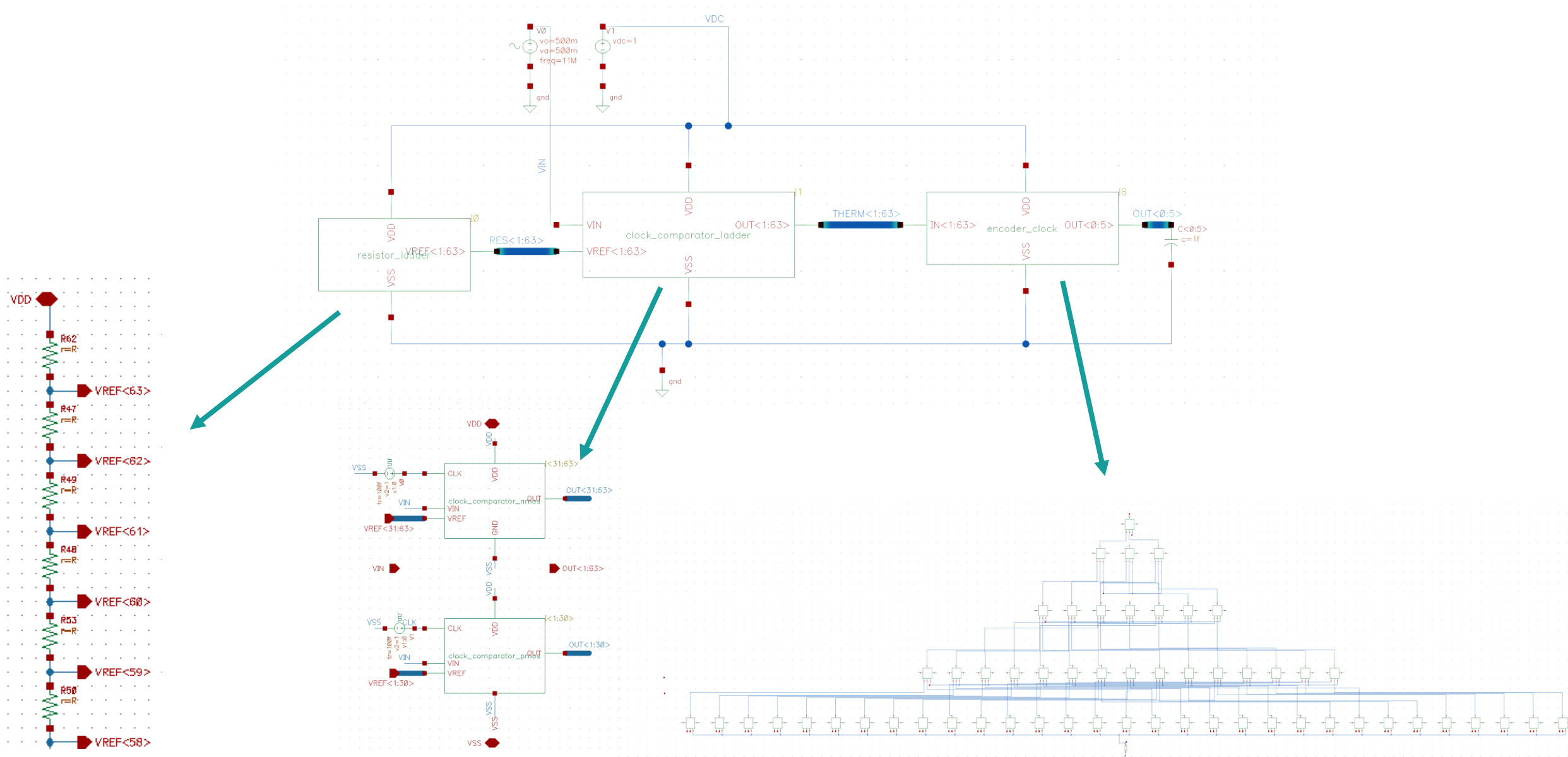
High-Speed, Low Power Dynamic 6-bit Flash ADC

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Top Level Schematic







Theoretical Analysis - Assumptions

- $f = 500MHz$
- $C_L = 1fF$
- $R_{NMOS} \approx 10k\Omega \times \mu m/W$
- $R_{PMOS} \approx 20k\Omega \times \mu m/W$
- $\frac{C}{L} = 1fF/\mu m$



Theoretical Analysis - Power

Resistor Ladder

$$P_{avg} = \frac{V^2}{R_{tot}}$$

$$V = 1V$$

$$R_{tot} = 63 \times 30\Omega = 1890\Omega$$

$$P_{avg} = \frac{1}{1890}W = 530\mu W$$

Comparator Ladder

Expected number of comparators to switch from one sample to the next is 21

$$P_{avg} = \sum \alpha_i C_i V_{DD}^2 f$$

Clock:

$$\alpha C = 1 \times (4fF + 1fF) = 5fF$$

StrongArm:

$$\alpha C = 2(0.5 \times (3fF + 8fF)) = 11fF$$

SR Latch:

$$\alpha C = 0.3 \times 6fF = 2fF$$

$$P_{avg} = 18fF \times 1V^2 \times 500MHz = 9\mu W$$

$$P_{avg_{tot}} = 63 \times 9\mu W = 567\mu W$$

Encoder

Expected number of muxes to switch from one sample to the next is 20

$$P_{avg} = \sum \alpha_i C_i V_{DD}^2 f$$

$$C = C_{diffusion} + C_{gates}$$

$$C_{gates} = 8 \times 0.12\mu M \times \frac{1fF}{m} = 0.96fF$$

$$C_{diff} = 12 \times 0.12\mu M \times \frac{1fF}{m} = 1.44\mu F$$

$$P_{avg} = 0.3 \times 2.4\mu F \times 1V^2 \times 500MHz = 0.4\mu W$$

$$P_{avg_{tot}} = 63 \times 0.4\mu W = 25.2\mu W$$

$$P_{avg} \approx 1.1mW$$



Theoretical Analysis - Speed

StrongArm Latch

$$t_{SAL} = 0.69RC$$

Based on the widths of the transistors:

$$\begin{aligned} R &= 2.5k\Omega + 5k\Omega + 10k\Omega \\ &= 17.5k\Omega \end{aligned}$$

Based on the output capacitance of the StrongARM latch:

$$C = 8fF$$

$$\begin{aligned} t_{SAL} &= 0.69 \times 17.5k\Omega \times 8fF \\ &= 96ps \end{aligned}$$

SR Latch

$$t_{SRL} = 0.69RC$$

Based on the widths of the PMOS transistors:

$$R = \frac{20k\Omega}{0.4} = 50k\Omega$$

$$C \approx 3.2fF$$

$$\begin{aligned} t_{SRL} &= 0.69 \times 50k\Omega \times 3.2fF \\ &= 110ps \end{aligned}$$

Multiplexer

$$t_{MUX} = 0.69RC$$

Since it uses minimum sized transistors

$$R_{invp} = \frac{20k\Omega}{0.12} = 167k\Omega$$

$$C = 0.12fF \times 8 + C_L$$

$$\begin{aligned} t_{MUX} &= 0.69 \times 167k\Omega \times 1.96fF \\ &= 225ps \end{aligned}$$

$$t_{worst} = t_{SAL} + t_{SRL} + 5t_{MUX} \approx 1.33ns$$

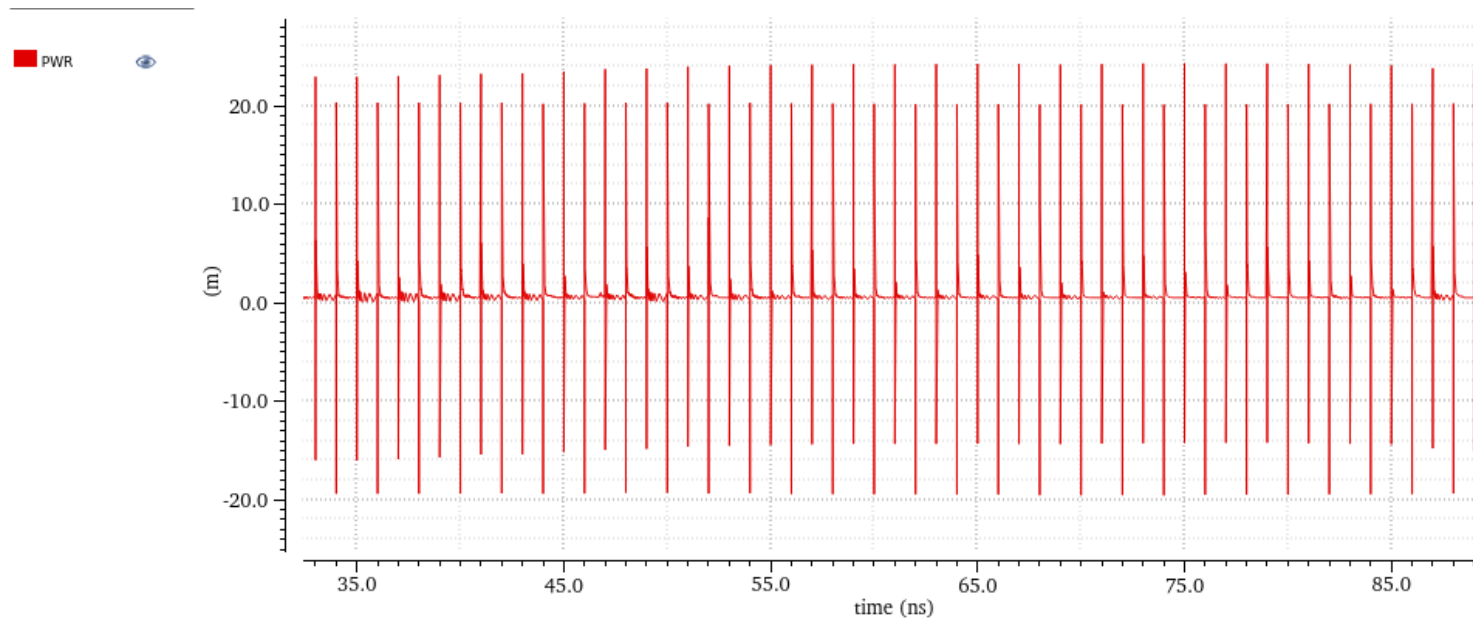
Results - Power

Average Power:

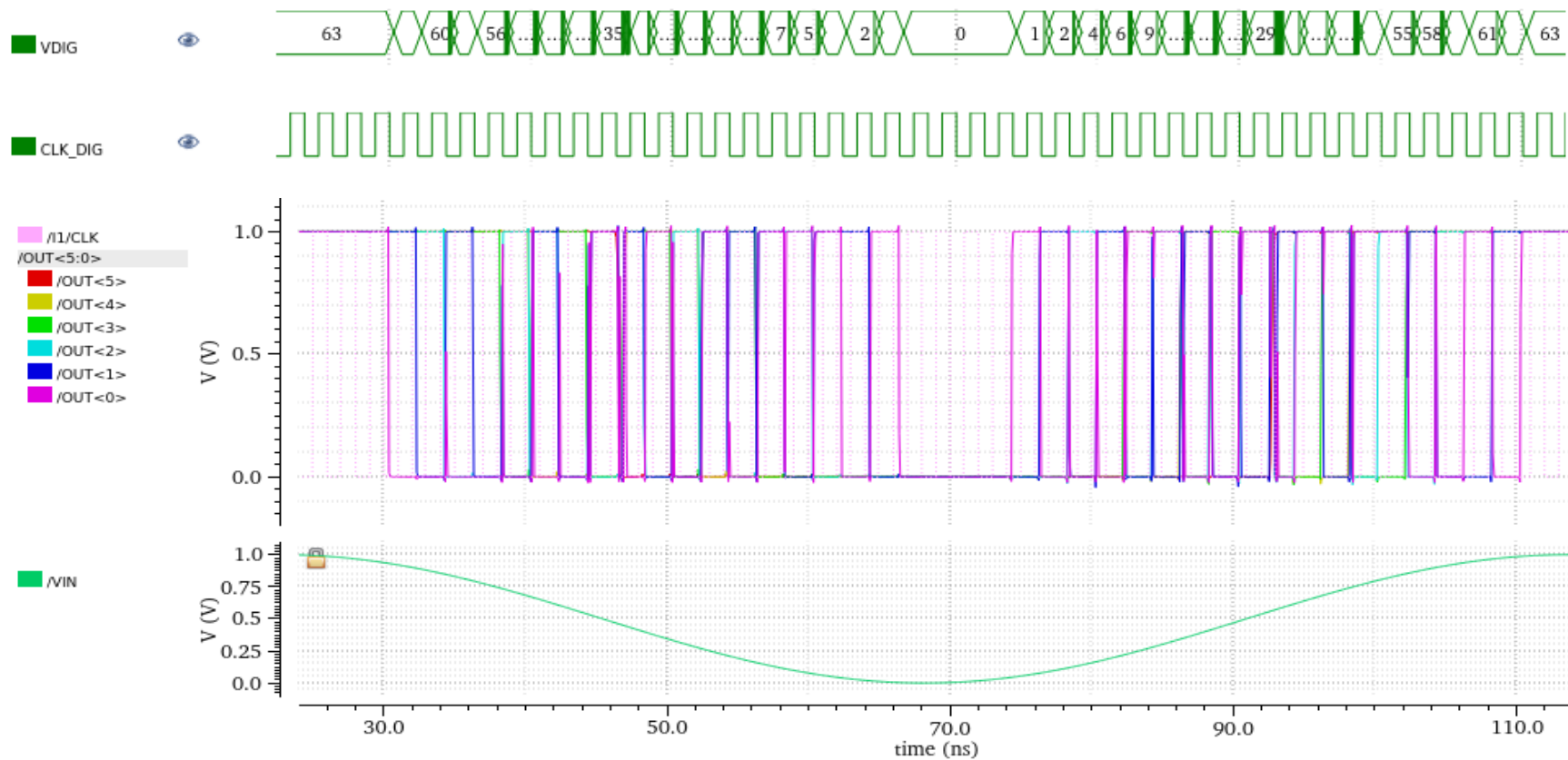
$806.9\mu W$

Max Instant Power:

$24mW$



Results - Accuracy



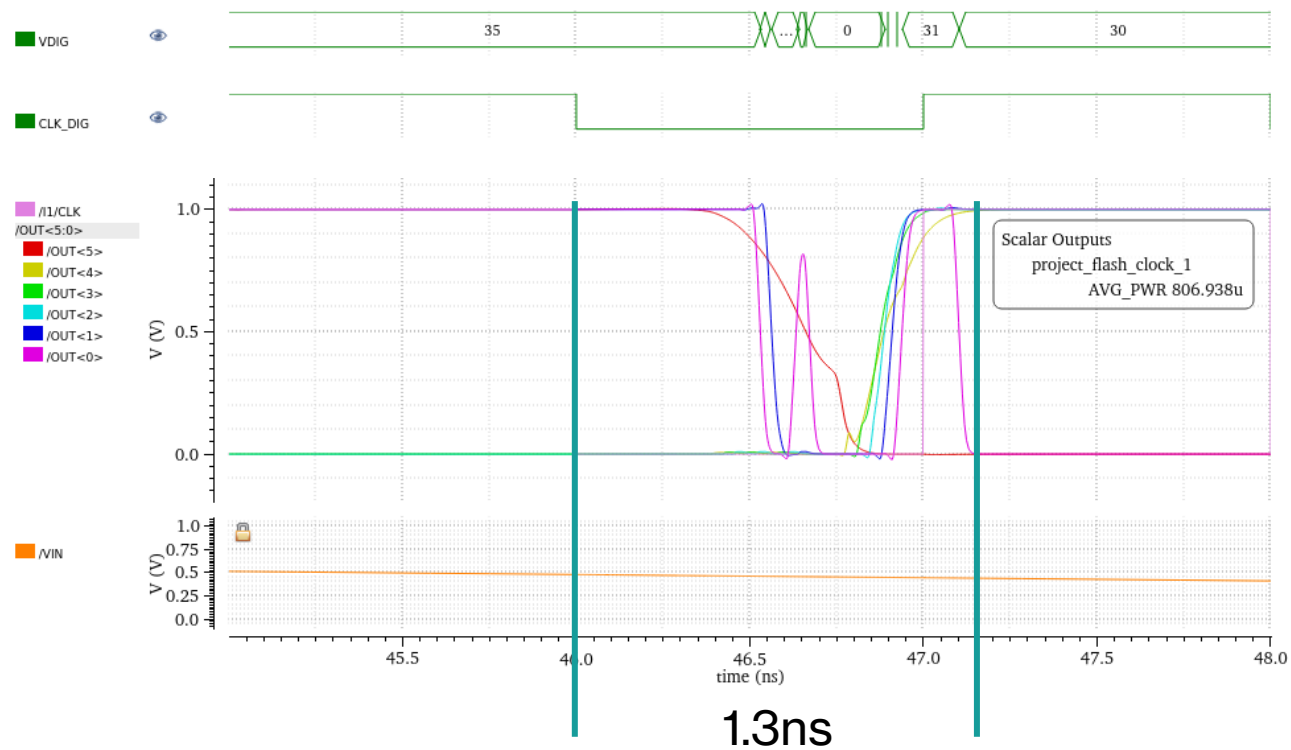
Results - Speed



Worst Case Latency:

1.3ns

0.7ns before next sample





Other Techniques - 45nm 5-bit Flash ADC

B. S. Pavan, et al., “Design of Flash ADC with CMOS Logic Using 45nm Technology”

- The proposed design is heavily based on this design, with the encoder being an almost identical replica of the one in this paper
- This design uses static comparators, forcing it into a tradeoff between power and speed.
- Achieved $6MS/s$ with an impressively low $2.7nW$ power consumption

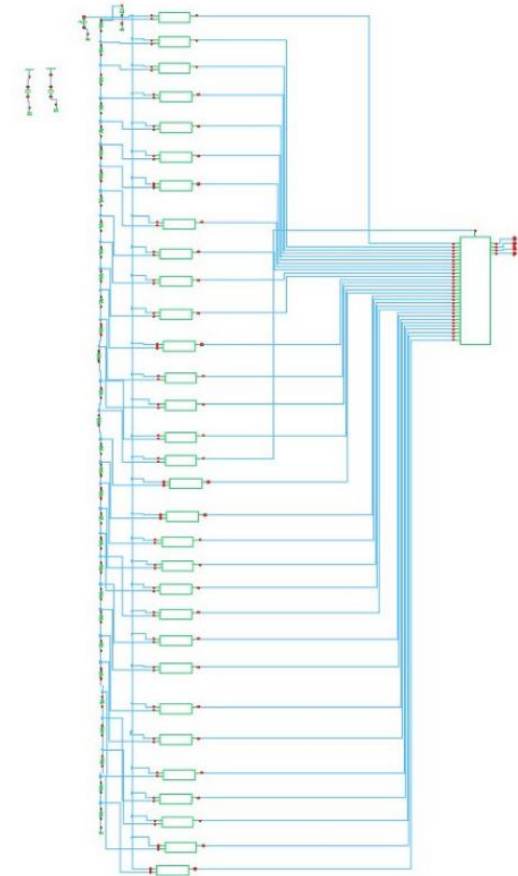


Fig. 10. Implementation of 5-bit Flash ADC using CMOS logic



Other Techniques – 180nm 6-bit Flash ADC

T. Tri Vo, et al. “A 6-bit Low-Power High-Speed Flash ADC using 180nm CMOS process”

- Uses a different technology branch, yet achieves a high speed low power
- Does not support rail-to-rail input voltages, working with $V_{DD}=1.8V$ and supporting an inputs in a 1V range
- Achieved $400MS/s$ with only $16mW$ while using 180nm technologies

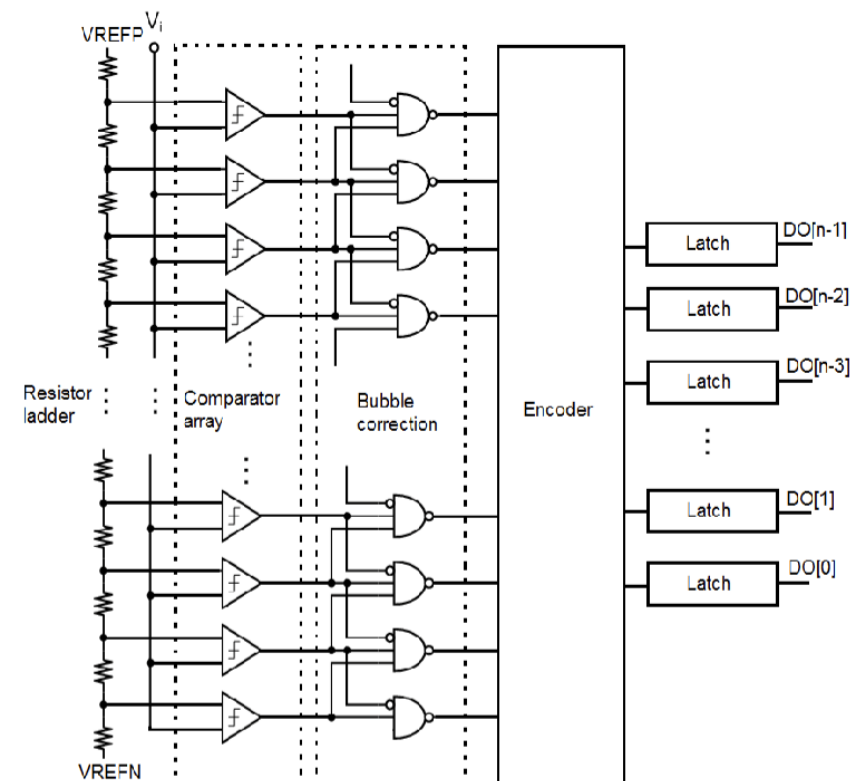


Fig. 1. The proposed Flash ADC architecture

Conclusion

- High-speed (500MS/s)
- Low Power ($\sim 800\mu W$)
- Rail-to-Rail voltage support
- Future Steps:
 - Scale multiplexers across layers to optimize speed based on load
 - Layout+ Layout Simulation
 - Frequency Based Analysis



References

- [1] B. S. Pavan, et al., “Design of Flash ADC with CMOS Logic Using 45nm Technology,” IEEE Xplore, 2024. [Online]. Available : <https://ieeexplore.ieee.org/document/10548132>
- [2] T. Tri Vo, et al. “A 6-bit Low-Power High-Speed Flash ADC using 180nm CMOS process,” Science & Technology Development, 2014.
- [3] A. Alshehri, et al., “StrongARM Latch Comparator Performance Enhancement by Implementing Clocked Forward Body Biasing,” 2018 25th IEEE International Conference on Electronics, Circuits and Systems (ICECS), 2018. [Online]. Available : <http://dx.doi.org/10.1109/icecs.2018.8617903>